University of St. Gallen

University of St.Gallen – School of Management, Economics, Law, Social Sciences, International Affairs and Computer Science

20 Years of the Double Pukelsheim System in Switzerland: What Is the Mechanical Effect of the Transition from the Hagenbach-Bischoff Method on the Seat Distribution of Smaller Parties?

Jan Moritz Beyn

Talackerstrasse 11

8274 Tägerwilen

+41 79 551 89 39

janmoritz.beyn@student.unisg.ch

23-611-908

University of St. Gallen

Bachelor Thesis Economics

Supervisor: Dr. Aurélien Sallin

Submission date: 17.11.2026

Abstract

Table of Contents

Abstract I
Introduction 1
Theory and Literature Review 3
    Theory 3
    Literature Review 5
Hypotheses 6
    Hypothesis 1 6
        Sub-hypothesis H1a: Differences between cantons 7
    Hypothesis 2 7
Analysis 7
    Data Cleaning 8
    Results 10
        Smaller-party representation across elections 10
        Relative smaller-party representation across elections 11
        H1a: Differences by Double Pukelsheim seat count 12
        Interpretation of the Results 13
Discussion 13
Conclusion 13
Bibliography 13
Appendix 18
    Appendix 1: Explanation Data Cleaning Process 18
    Appendix 2: Code for Data Cleaning 18
    Declaration of AI-Assistance 18

Introduction

On the 18th of December 2002 the Swiss Federal Supreme Court (FSC) came to a conclusion, which disrupted the Swiss Electoral System. In BGE 129 | 185 the FSC criticized that in the elections for the Council of the City of Zurich, the thresholds for parties to win a seat, vary heavily from district to district. Through that not every voter has the same influence and therefore is the Principle of Equal Treatment (Gleichhandlungsgebot) not given anymore. Based on this criticism the court decided that the City of Zurich needs to implement corrective measures, it let the City decide which these would be. (BGE 129 I 185, E. 7.3, S. 199–200) The Government of the Canton of Zurich decided to keep the consituency structure (Regierungsrat des Kantons Zürich, 2014, pp. 3–4). Instead the canton of Zurich comissioned the german mathematician Friedrich Pukelsheim to develop a comprehensive and at the same time minimally invasive solution for the problem. Eventually Pukelsheim came up with a solution named "biproportional divisor method with standard rounding" which is today often simply called "double Pukelsheim" or "Biproportional apportionment". (Lüscher, 2024, pp. 9-10)

The main differentiation of the Pukelsheimer Proporz, is that it through the so called upper-apportionment artificially creates a single electoral district for the initial allocation of the parliamentary seats to the parties or list groups. Through this mechanism, the difference between Seat- and voteshares gets minimized and with that also the threshold for parties to enter the parliament gets lowered. (Lüscher, 2023, p. 575)

Through it's upper-apportionment mechanism and the lowered threshold, the reform is especially relevant for the representation of smaller parties. Since votes that would previously have been ineffective in small constituencies can now contribute to a seat gain at the cantonal level. Sandro Lüscher illustrates this mechanical effect with the case of Schwyz in 2016: under the old Hagenbach-Bischoff system, the Green Liberal Party would not have won a single seat, whereas under the new method they obtained three seats (Lüscher, 2025, p. 1-3).

Since the implementation through the Canton of Zurich in 2007, nine more cantons followed till date: Schaffhausen (2008), Aargau (2009), Nidwalden (2014), Zug (2014), Schwyz (2016) , Valais (2017) , Uri (2020), Graubünden (2022) in 2025 also the voters from Basel Landschaft decided to implement the Biproportional apportionment from 2027 on (Lüscher , 2024, p.17 ; Kanton Basel-Landschaft, 2025a, p.7 ; Kanton Basel-Landschaft, 2025b) While these cantons already implemented the algorithm, more cantons are currently discussing the implementation too. So is in Thurgau currently the initiative "Mehr Transparenz bei Wahlen im Thurgau" pending at the Grand Council of the Canton and special parliamentary comittee is currently examining the initiative, also in the canton of Lucerne, the iniatitive "Faire Wahlen ohne Listenflut" was submitted and the voters will be able to decide on the effectuation (Grosser Rat des Kantons Thurgau, n.d.; Regierungsrat des Kantons Luzern, 2026, pp. 19–20). In the canton of St.Gallen in 2026 an initiative to inaugurate the algorithm was launched (Initiativkomitee «Damit jede Stimme zählt», n.d.).

The amount of cantons who have already implemented the system or are debating about doing so, shows that the decision by the FSC had an substantial impact on how Swiss Citizens elect their regional parliaments. It seems that the double-Pukelsheimer through it's wide implementation across cantons is a success story. Since the first election under double Pukelsheim

in Zurich in 2007, the system has been used in 32 cantonal parliamentary elections.[1]

Taken together, this number of elections and almost 20 years of experience provide a reason to evaluate the impact of the reform, especially on smaller parties. Has the algorithm kept its promises, and how large is the actual difference in seat allocation between the Hagenbach-Bischoff and Double Pukelsheim systems? Are there other major differences, for example in how the reform performs in smaller versus larger parliaments, and does the reform lead to a systematic increase in party-system fragmentation?

The central research gap lies in the fact that the existing literature mainly discusses the Doppelproporz in legal, normative, or system-level terms. It emphasizes improved proportionality, fewer weightless votes, and better opportunities for smaller parties, while also pointing to trade-offs such as territorial distortions and counterintuitive seat allocations (Lüscher, 2023, 2024, 2025). The closest precedents for the calculation used in this thesis are the counterfactual comparisons by Pukelsheim and Schuhmacher, Lüscher, and Leuzinger. Pukelsheim and Schuhmacher use the actual 2011 Zurich election under Doppelproporz combined with hypothetical list alliances to calculate how the same votes would have been translated into seats under the former Hagenbach-Bischoff system, showing that larger parties receive more seats at the expense of several smaller parties under the old rule (Pukelsheim & Schuhmacher, 2011, Section D, pp. 1590–1592, Table 2). Lüscher performs a similar comparison for the 2016 Schwyz election (Lüscher, 2024, p. 13, Table 3). Leuzinger compared the then most recent elections in the seven cantons that had adopted Doppelproporz by 2018 with hypothetical allocations under their former electoral systems (Leuzinger, 2018). Unlike the present study, his counterfactual assumes that the list alliances observed in the final pre-reform election would have persisted. The two studies therefore produce different results for some elections. By contrast, Lüscher's principal empirical analysis follows Swiss cantons over time and estimates how outcomes such as disproportionality and party-system fragmentation would probably have developed in the reforming cantons had they not adopted Doppelproporz (Lüscher, 2024, pp. 126–143). As discussed further in the literature review, the remaining gap is therefore the absence of a standardized application across multiple elections and cantons that measures the resulting seat gains of smaller parties without imposing assumptions about unobserved post-reform list alliances.

Against this background, this thesis examines the mechanical effect of the electoral reform on the parliamentary representation of smaller parties. More specifically, it asks and shows statistically how many seats smaller parties gain through the transition from the Hagenbach-Bischoff system to Double Pukelsheim in Swiss cantonal parliamentary elections. It further examines whether the reform performs differently in smaller and larger parliaments and whether it leads to a systematic increase in party-system fragmentation. Its contribution is therefore not the invention of the counterfactual calculation, but its systematic application

[1] The 32 elections comprise Zurich (2007, 2011, 2015, 2019, and 2023; Amt für Statistik und Daten Kanton Zürich, 2026); Schaffhausen (2008, 2012, 2016, 2020, and 2024; Kanton Schaffhausen, n.d.; Staatskanzlei Kanton Schaffhausen, 2020, 2024); Aargau (2009, 2012, 2016, 2020, and 2024; Department für Finanzen und Ressourcen des Kantons Aargau, 2009; Regierungsrat des Kantons Aargau, 2013; Statistisches Amt Kanton Aargau, n.d.); Nidwalden (2014, 2018, 2022, and 2026; Staatskanzlei Nidwalden, 2014, 2018, 2022, 2026); Zug (2014, 2018, and 2022; Kanton Zug, 2022a, 2022b, 2022c); Schwyz (2016, 2020, and 2024; Bundesgericht, 2012); Valais (2017, 2021, and 2025; Kanton Wallis, 2016, 2020, 2024; Staat Wallis, 2017); Uri (2020 and 2024; Standeskanzlei Uri, 2024a, 2024b); and Graubünden (2022 and 2026; Kanton Graubünden, 2022, 2026).

across 32 cantonal elections using a consistent definition of smaller parties and aggregated seat-gain measures.

Theory and Literature Review

Theory

The following section will provide an overview over the usage of different electoral systems throughout the time within Switzerland. It aims to show why the debate about which electoral system to use is one held often and repeatedly. For that it presents the underlying arguments brought upfront to change or keep a system. The structure of this section is the following: There is a short explanation of the historical situation in Switzerland, secondly the current situation will be explained and thirdly the transition from Hagenbach-Bischoff to Dobule Pukelsheimer in the corresponding cantons will be examined.

Till the end of the 19th century swiss voters elected the regional parliaments and great councils only through Majority vote. In a majority voting system, the candidates or parties who receive the relative or absolute majority receive the seats, therefore this systematically favours the dominating parties. The disadvantages for smaller parties lead to the urge of some of the political minorities, to change to a proportional system. In 1891 the canton of Ticino, was the first one to use it to vote for their Grand Council, till the end of World War I , 7 1/2 more cantons followed. When the socialist and conservative minorities on their third try in 1918 achieved, that the National Council will be elected through a proportional system, it boosted the general adoption of it across Switzerland. Most of the cantons Grand Councils as aswell the Federal Council where elected with the Hagenbach-Bischoff proportional system. In the beginning of the 21th century, the cantons of Graubünden and Appenzell Innerhoden were the only ones to still vote their parliaments with the majority system. (Gilg, 2014, "Mehrheits- und Verhältniswahl" section ; Bundeskanzlei, 2013, p. 2) Nowadays the Kanton Appenzell Innerhoden is the only remaining canton still voting the Grand Council with the majority system (Bundeskanzlei, 2025, p. 18). It can therefore be said, that the proportional system is the dominating electoral system for regional parliaments in Switzerland.

While earlier the Hagenbach-Bischoff proportional system, was the dominating form. This changed over the last 20 years since the development of the Double Pukelsheimer. As of 2025, eleven cantons distribute the parliament seats with Hagenbach-Bischoff and ten cantons distribute them with the DP, two cantons namely Glarus and Basel-Landschaft use the Sainte-Laguë method and two namely Ticino and Vaude use the Hare/Niemeyer method (Bundeskanzlei, 2025, p.18). Among the cantons analyzed here, Hagenbach-Bischoff was the former allocation method in nine of the cantons in at least part of the territory except in Graubünden, which previously used Majorz throughout. Zug's Neuheim constituency and Uri's smaller municipalities also used Majorz rather than Hagenbach-Bischoff (Bundeskanzlei, 2013, Table 1; Bundesgericht, 2010b, E. 5.1; Standeskanzlei Uri, 2020, p. 1; Statistisches Amt des Kantons Zürich, 2011, p. 286; Regierungsrat des Kantons Schaffhausen, 2007, p. 6; Regierungsrat des Kantons Aargau, 2007, p. 18).

Followingly the arguments brought up in each reform canton will be discussed in chronological order.

The year in the beginning of each line, marks the year of first application of the double Pukelsheimer in the corresponding canton.

- **2007:** Zurich became the first canton to use Double Pukelsheim, which it had commissioned Friedrich Pukelsheim to develop. The reform followed a 2002 voting-rights complaint about elections to Zurich's city parliament. The Swiss Federal Supreme Court partially upheld the complaint, finding that the thresholds in some districts were too high. The canton of Zurich implemented the Pukelsheimer Proporz then with a 5% local (district) for parties to enter the parliament. (Lüscher, 2024, pp. 9–10, 33-36) In 2021, Zurich added a 3% cantonwide threshold, first applied in the 2023 election. A change to how that threshold is calculated was adopted in 2025 and will first apply in the 2027 election (Regierungsrat des Kantons Zürich, 2024, pp. 2, 5; Regierungsrat des Kantons Zürich, 2025, p. 1).
- **2008:** Schaffhausen : In a 2004 popular vote, Schaffhausen's electorate approved a reduction in the cantonal parliament from 80 to 60 seats. Shortly after a motion was submitted which wanted to reform the electoral law to better protect smaller parties. After this the Schaffhausen Government proposed a reform. Subsequently the parliament (2007) and the voters (2008) approved the changes. In 2015, the cantonal parliament added a majority clause, this ensures that the stronges party receives at least one seat in the corresponding district. (Lüscher , 2024 , pp. 41- 43)
- **2009:** Aargau's voters voted in the year 2003 for a popular vote, which wanted to lower the amount of parliament seats from 200 to 140. This decision naturally highered the tresholds in the districts and therefore a second constitutional amendment allowing to merge small districts was proposed and approved by the voters. Though the parlamentarians decided against making use of it. Therefore the electoral equity wasn't given anymore. The Government of Aargau had to act and proposed to the Grand Council to either merge districts or to change to the Double-Pukelsheim. The Grand Council again decided to not change anything. Shortly after this decision by the Grand Council the FSC issued in a verdict, that there are existing inequalities, because of the varying thresholds and district sizes. The court ordered that this needs to be changed. In Septembere 2007, the Grand Council with a thin majority then decided to implement the Double-Pukelsheimer. In 2011 a 5% local (district) and a 3% global quorum was added after approval of the Grand Council and the voters (Lüscher, 2024, pp. 38 - 40)
- **2014:** Nidwalden: In the year 2010 members of the Green Party in Nidwalden filed a complaint regarding the seat allocation process for the cantonal elections. That complaint got rejected initially by the Cantonal Court but later the cantonal decision got overturned by the FSC. Since the FSC didn't approve the claim, that the allocation process was justified historically. As a reaction on this decision the parliament of Nidwalden came up with multiple possible reforms. The most popular of the suggested reforms, was the change towards the biproportional system. The parliament and the electorates approved followingly the implementation of the Double-Pukelsheimer. (Lüscher, 2024, p. 44- 45 ; Bundesgericht, 2010a, p.353-364)
- **2014:** Zug: In 2010, a group around multiple parties formed a complaint at the FSC regarding the differently sized districts. The Federal Supreme Court allowed the appeals main point. As a reaction to this decision the cantonal Government present three different possibilities on how to adapt to the FSC ruling. One of them and the favoured by the Government was

the inauguration of the DP. The Parliament added based on the model of Aargau a 5% local and 3% global threshold. A competing proposal to retain the existing system was struck down by the FSC in 2013. Voters approved the biproportional reform on 22 September 2013. (Lüscher, 2024, p. 47-51 ; Bundesgericht, 2013, E. 4, pp. 204–205 ; Bundesgericht, 2010b, E. 4.5, pp. 378, 385)

- **2016:** Schwyz
- **2017:** Valais
- **2020:** Uri
- **2022:** Graubünden
- **2027:** Basel-Landschaft

Literature Review

The analytical approaches which come closest to the the method used in this thesis, are the in the Introduction named researches by Leuzinger (2018), Pukelsheim and Schuhmacher (2011, Section D , pp. 1590 –1592) aswell as the Doctoral Thesis from Lüscher (2024). Pukelsheim and Schuhmacher use the observed 2011 Zurich election under Doppelproporz to calculate how the same votes would have been translated into seats under the former Hagenbach-Bischoff system; their comparison shows seats moving from several smaller parties to larger parties under the old rule. In their calculations, Pukelsheim and Schuhmacher make two things differently, than this thesis does. Firstly they assume that there is under Doppelproporz no 5% threshold within the electoral districts, secondly do they form hypothetical list alliances based on their political likeliness and former list alliances. This thesis though implements the 5% threshold for the canton of Zurich and all other known specifications of cantons, but it doesn't carry forward or hypothetically creates new list alliances, because whether parties would have renewed them is unkown. Leuzinger provides the closest multicanton precedent. He compares the most recent Doppelproporz election in each of seven cantons with a counterfactual allocation under the former electoral system and finds that, on average, approximately 10% of the seats would have been assigned to a different party. His counterfactual differs from the one used here in an important respect. For Zurich, Schaffhausen, and Aargau, where the former Hagenbach-Bischoff system allowed list alliances, Leuzinger assumes that parties would have repeated the alliances formed in the last pre-reform election. (Leuzinger, 2018, pp. 41-42)

To sum up, this study does not follow Leuzinger's or Pukelsheims and Schuhmachers approach of carrying historical list alliances forward into the counterfactual allocation. List alliances are in this thesis seen as strategic and election specific rather than fixed institutional parameters. Their formation depends on the parties contesting an election, their relative electoral strength, ideological proximity, district-level incentives, and expectations about possible seat gains (SOURCE?). In Zurich, for example, the last Kantonsrat election under Hagenbach–Bischoff took place in 2003. Assuming that the alliances formed in 2003 would still have applied in 2023 despite two decades of changes in the party system would therefore introduce a strong and unverifiable behavioural assumption. For example did parties who have formed list-alliances in the canton of Zurich in 2003 not exist or run for office anymore in 2023, other parties which ran in 2023 on the other hand like the Green Liberal Party (GLP) did not exist yet in 2003 (Statistisches Amt des Kantons Zürich 2003, table 3; Statistisches Amt des Kantons Zürich 2026; Grünliberale Partei Kanton Zürich n.d.). The analysis instead includes alliances only

when they were actually registered in the election under examination. This provides a transparent counterfactual that holds the observed electoral configuration constant and isolates the mechanical consequences of applying Hagenbach–Bischoff. List alliances pool the votes of connected lists for the first seat-allocation stage and can therefore affect which lists win seats (Pukelsheim & Schuhmacher, 2011, p. 1590).

Lüscher similarly compares the actual allocation in the 2016 Schwyz election with a simplified Hagenbach-Bischoff counterfactual, finding that the dominant SVP and CVP benefit under Hagenbach-Bischoff, whereas the left-wing alliance and particularly the GLP benefit from Doppelproporz in that single election (Lüscher, 2024 , p.13).

The choice regarding the list alliances does not imply that parties would have contested the election without alliances under Hagenbach–Bischoff. Rather, their counterfactual alliance decisions cannot be reliably observed or reconstructed. Because alliances can mitigate the disadvantages experienced by smaller parties under Hagenbach–Bischoff (Pukelsheim & Schuhmacher, 2011, p. 1590), excluding unobserved alliances may affect the estimated magnitude of the resulting seat changes.

Hypotheses

To be able to answer the research question: 20 Years of Double Pukelsheim in Switzerland: What Is the Mechanical Effect of the Transition from Hagenbach-Bischoff on the Seat Allocation of Smaller Parties and Party-System Fragmentation?

We need to formulate hypotheses. The hypotheses are based on the theoretical considerations and the literature review. The first hypothesis focuses on the mechanical effect of the electoral reform on smaller parties, while the second hypothesis addresses the potential impact of the reform on party-system fragmentation.

Hypothesis 1

For each election e, smaller parties are those with less than 10% of the vote in the analyzed area: the whole canton except in Uri, where only the four municipalities using Double Pukelsheim are included. Let S_e^{DP} and S_e^{HB} be their combined seats under Double Pukelsheim and the counterfactual Hagenbach-Bischoff allocation, respectively, and let M_e be the number of seats allocated through Double Pukelsheim and included in the comparison. Their gain in seat share, shown for each election in Figure 2, is

$$\Delta_e = 100\frac{S_e^{DP} - S_e^{HB}}{M_e} \quad \text{percentage points} .$$

The main hypothesis concerns the unweighted mean of these election-level gains, $\bar{\Delta} = E^{-1}\sum_{e=1}^{E}\Delta_e$, across the E analyzed elections. Thus, "more than 5% more seats" means a mean gain greater than **five percentage points of the seats included in the comparison**, rather than a 5% increase relative to the Hagenbach-Bischoff seat total.

H_0 (null hypothesis): The mean gain in smaller parties' seat share under Double Pukelsheim is at most five percentage points: $\bar{\Delta} \leq 5$.

H_1 (main hypothesis): The mean gain in smaller parties' seat share under Double Pukelsheim exceeds five percentage points: $\bar{\Delta} > 5$.

The 10% threshold is based on that the Swiss Federal Supreme Court has argued that direct and natural quorums, understood as barriers to representation within an electoral district, should generally not exceed 10% (BGE 131 I 74, E. 5.3–5.5; Bundeskanzlei, 2013, p. 8).

Sub-hypothesis H1a: Differences between cantons
The gain in smaller parties' seat share may vary across cantons. As an exploratory comparison, cantons where more than 100 seats are allocated through Double Pukelsheim (ZH, AG, and VS) are compared with cantons where at most 100 seats are allocated through Double Pukelsheim (UR, NW, ZG, SH, and SZ).[2] Graubünden is shown separately because its pre-reform system was Majorz and its HB allocation is hypothetical (Bundeskanzlei, 2013, Table 1).

H_0a: The two groups have the same mean canton-level gain in smaller parties' seat share under Double Pukelsheim.

H_1a: The mean canton-level gain in smaller parties' seat share under Double Pukelsheim differs between the two groups.

For each canton, the election-level gains Δ_e are averaged first; the group mean then averages these canton means so that cantons with more observed elections receive no extra weight. The difference is described in percentage points without a directional prediction or a claim that canton size causes it. Former district magnitudes, legal thresholds, party competition, and canton-specific allocation rules could also shape the result.

Hypothesis 2

$H_0$2 (null hypothesis): The transition to the double Pukelsheim system did not lead to a systematic increase in party-system fragmentation.

$$H_{0,2}: \quad \mathrm{ENP}_{\mathrm{post}} - \mathrm{ENP}_{\mathrm{pre}} \leq 0$$

$H_1$2: The transition to the double Pukelsheim system led to a systematic increase in party-system fragmentation.

$$H_{1,2}: \quad \mathrm{ENP}_{\mathrm{post}} - \mathrm{ENP}_{\mathrm{pre}} > 0$$

The effective number of parties is calculated using the Laakso–Taagepera index (Laakso & Taagepera, 1979):

$$\mathrm{ENP}_v = \frac{1}{\sum_{i=1}^{n} v_i^2} \qquad \mathrm{ENP}_s = \frac{1}{\sum_{i=1}^{n} s_i^2}$$

v = votes shares, s = seat shares

Analysis

H1: The analysis takes the actual seat allocation under the Pukelsheimer Proporz as its starting point. Based on the available election data at the municipal or district level, it then reconstructs how many seats each party would have obtained under the former Hagenbach-Bischoff system. The comparison between the actual seat distribution under the new system

[2] Official seat totals: ZH (Amt für Statistik und Daten Kanton Zürich, 2026); AG (Statistisches Amt Kanton Aargau, n.d.); VS (Kanton Wallis, 2024); UR (Standeskanzlei Uri, 2024b); NW (Staatskanzlei Nidwalden, 2026); ZG (Kanton Zug, 2022c); SH (Staatskanzlei Kanton Schaffhausen, 2024); SZ (Kanton Schwyz, 2024); GR, shown for context (Kanton Graubünden, 2026).

and the hypothetical distribution under the old system makes it possible to identify how many seats each party gained or lost as a result of the reform. In this way, the study assesses the mechanical effect of the electoral-system change on party representation, with a particular focus on smaller parties.

For the purposes of this thesis, a smaller party is defined as an electoral list or list group with less than 10% of the vote share in the analyzed area: the whole canton except in Uri, where the vote shares cover only the four municipalities using Double Pukelsheim. Youth lists and other special-interest lists, such as women's lists or pensioners' lists, are treated as separate analytical units whenever they are reported separately in the official election results; their votes and seats are not attributed to a mother party. Accordingly, the term *party* in the analysis refers to the standardized electoral list or list group recorded in the dataset and does not necessarily correspond to a broader organizational party family. The choice of 10% as the threshold is grounded in both legal and normative considerations. In its decision BGE 131 I 74, the Swiss Federal Supreme Court argued that direct and natural quorums, understood as barriers to representation within an electoral district, should generally not exceed 10% (E. 5.3–5.5; see also Bundeskanzlei, 2013, p. 8). Using vote share rather than absolute party size is particularly appropriate because a party may be electorally small in one region while being comparatively strong in another. Low thresholds are closely linked to the principle of Erfolgswertgleichheit, or equality of vote success value, which requires that votes should be translated into seats as equally and effectively as possible (BGE 131 I 74, E. 3.1; Bundeskanzlei, 2013, pp. 8–9). Investigating the threshold is especially important, since none of the reform cantons had a <10% threshold in all districts before the implantation of the new electoral reform (Lüscher, 2024, p. 115).

Data Cleaning

(STILL NEED TO CLARIFY ON HOW MUCH OF DATA CLEANING IS EXPLAINED HERE AND HOW MUCH ONLY IN APPENDIX)

To be able to analyze the data from the different elections and compare the outcomes. each dataset has to be cleaned. The data For some cantons e.g. Valais and Schwyz the data is very scattered, therefore for there multiple datasets per election have to be cleaned and merged.

The procedure for the data cleaning was the following: For each canton in alphabetically order, for each election the data was cleaned.

To ensure the compatibility of the different datasets, the column names were standardized. The column names were renamed for each dataset to the following: "total_votes", "svp_votes", "sp_votes", "cvp_votes", "fdp_votes", "glp_votes", "bdp_votes", "green_votes", "other_votes". It might be that some of the datasets do not contain all of these columns, since the party landscapes varies geographically and over time.

For the counterfactual, the observed district-level party votes and district seat magnitudes are retained. Where Hagenbach-Bischoff was used before the reform, its allocation is reconstructed under the relevant former cantonal rules, including the former Zurich Wahlgesetz (§§ 87–88b), the pre-reform Aargau legislation, the former Uri Proporzgesetz (Arts. 22–23), and the former Valais law (Arts. 154–156); the canton-specific scripts document the remaining cases (Kanton Zürich, 1983/2005; Kanton Aargau, 2004; Kanton Uri, 2018; Kanton Wallis, 2004a). The HB allocations in Graubünden and in Zug's Neuheim constituency are

hypothetical comparisons rather than reconstructions of their former Majorz rules. The main counterfactual treats every party independently and does not carry historical list alliances forward. This is a deliberate identification choice: the reforming cantons' Double Pukelsheim designs did not provide for inter-party list alliances (Lüscher, 2024, p. 114 and Appendix A, Table A.1), so the alliances that parties would have formed in subsequent elections under the former rules are unobserved. Leuzinger (2018, p. 42), by contrast, assumes that the alliances from the last pre-reform election would have persisted. Vote shares for the analyzed area are derived from the available official *Wählerzahl* or reported vote-share figures; where necessary, the cleaning scripts reconstruct voter numbers from district-level party votes and seat magnitudes. The calculation and rounding procedures vary by canton and source file.

Schaffhausen reduced its parliament from 80 to 60 seats for the first Double Pukelsheim election. The counterfactual holds the later election's 60-seat district structure fixed, so it isolates the allocation method under that structure rather than reproducing every feature of the 2004 election (Regierungsrat des Kantons Schaffhausen, 2007, pp. 1, 6).

The official Double Pukelsheim results (`seats_dp`) are sourced from official cantonal authorities and are documented in the Bibliography. A single lookup table consolidates all official results for easy merging with the Hagenbach-Bischoff counterfactuals. All data processing is implemented in canton-specific R scripts (R/canton_*.R) which apply shared functions for algorithm execution and validation.

Results

Smaller-party representation across elections

> **i** Scope of the canton comparisons
>
> **Graubünden.** The canton used Majorz throughout before adopting Double Pukelsheim. Its HB allocation is hypothetical, so the observations are shown in grey and excluded from the main transition estimate (Bundeskanzlei, 2013, Table 1, p. 14).
>
> **Zug.** Neuheim used Majorz before the reform, while the other ten municipalities used proportional allocation. Neuheim participates in the observed canton-wide Double Pukelsheim allocation, so all 80 seats are retained. The HB allocation for Neuheim is a hypothetical uniform-HB comparison, not a reconstruction of its former Majorz rule (Bundesgericht, 2010b, E. 5.1; Kanton Zug, 2022a, 2022b, 2022c).
>
> **Uri.** Only Altdorf, Bürglen, Erstfeld, and Schattdorf used Double Pukelsheim in the elections analyzed here. They account for 37 of the Landrat's 64 seats; the remaining municipalities used Majorz and are excluded from both allocations in this comparison (Standeskanzlei Uri, 2020, p. 1). Uri's vote shares, including the less-than-10% classification of smaller parties, are based only on these four municipalities, not on the whole canton.
>
> **Valais.** Upper apportionment takes place separately within six electoral regions. If e.g. SVP/UDC stood in all six, it would enter six independent regional allocations; votes in one region cannot increase its seat entitlement in another. This analysis sums the resulting regional seats under one canton-wide party label and aggregates voter numbers to calculate canton-wide vote shares. The single party row and the classification of parties below 10% are therefore measures for comparison, not the regional units used to allocate seats (Lüscher, 2024, p. 57).

Figure 1 compares the observed Double Pukelsheim allocation with the counterfactual Hagenbach-Bischoff allocation for every cleaned election. For each election, the figure aggregates the seats and vote shares in the analyzed area of all parties below the 10% threshold defined above. The grey line connects the two allocations derived from the same election.

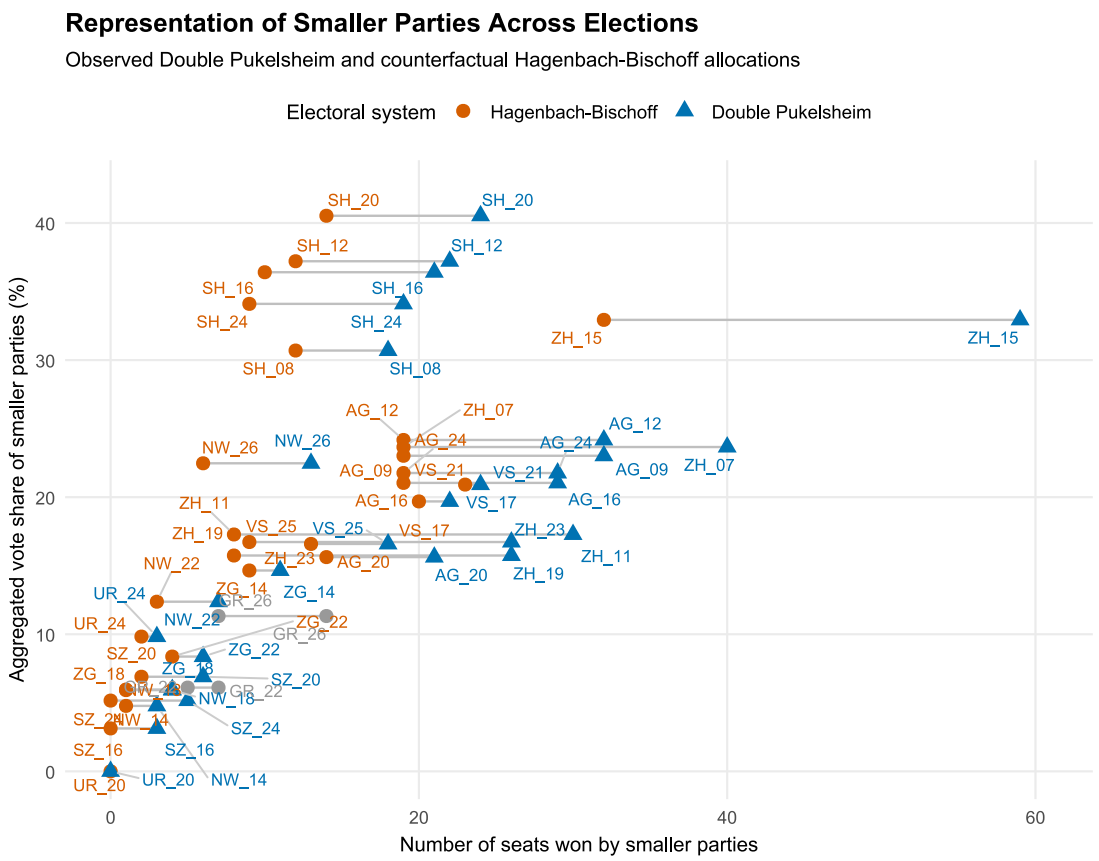


Figure 1: Seats won by smaller parties under Double Pukelsheim and the counterfactual Hagenbach-Bischoff allocation. Smaller parties have less than 10% of the vote share in the analyzed area.

Relative smaller-party representation across elections

Figure 2 shows the same elections using the percentage of analyzed Double Pukelsheim seats won by smaller parties on the x-axis. Each label reports the gain under Double Pukelsheim as a percentage-point share of the seats included in the comparison.

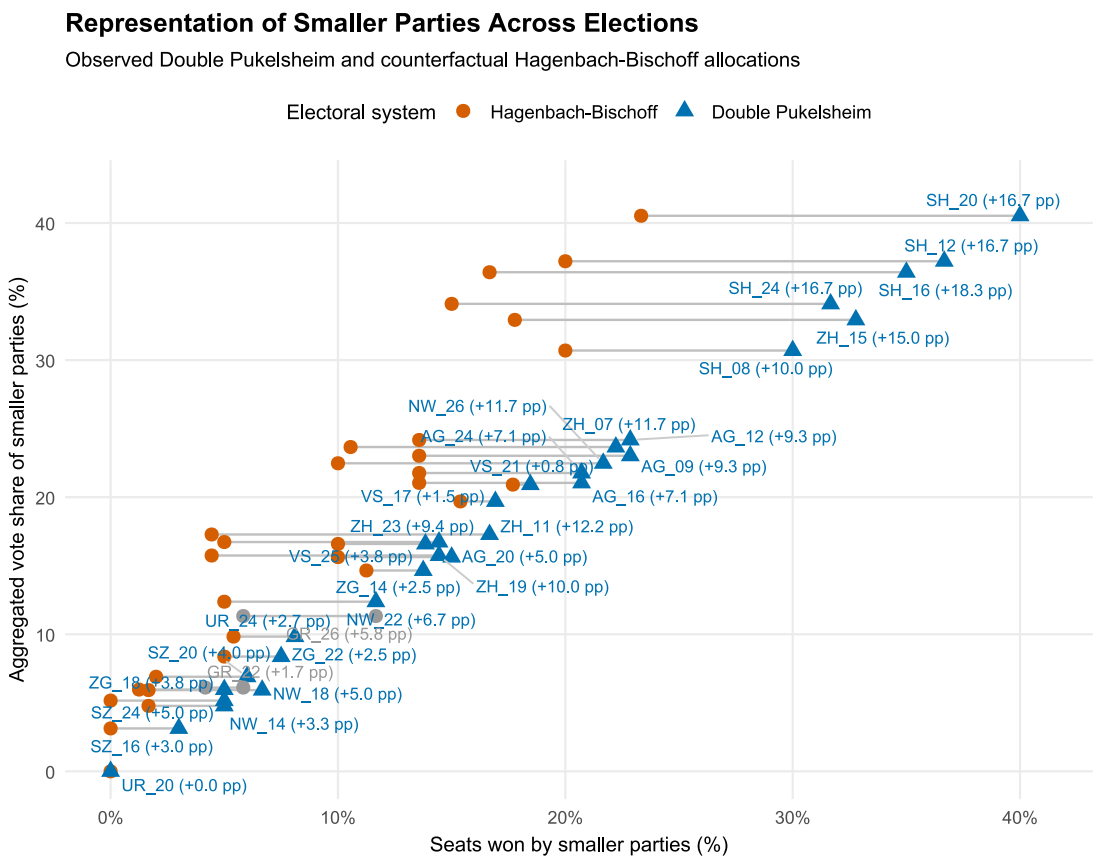


Figure 2: Percentage of analyzed seats won by smaller parties under Double Pukelsheim and counterfactual Hagenbach-Bischoff.

> **i** Hypothesis 1 summary
>
> Across all 30 elections (Graubünden excluded), the unweighted mean gain in smaller parties' share of analyzed seats was 7.7 percentage points (Q1: 3.4; median: 6.9; Q3: 11.2). The mean weighted by each election's number of analyzed seats was 7.9 percentage points. 16 individual elections exceeded five percentage points.

H1a: Differences by Double Pukelsheim seat count

The comparison below uses the same election-level percentage-point gain as H1. It averages elections within each canton before comparing the two groups defined by the number of seats allocated through Double Pukelsheim. Graubünden is only shown for context and excluded from the group means for the same reason as in the main H1 estimate.

Table 1: Mean gain in smaller parties' share of analyzed seats by canton (percentage points). Source: author's aggregation of the official cantonal election datasets.

Canton	Group	Elections	DP seats (parliament)	Mean gain (pp)
ZH	More than 100 DP seats	5	180 (180)	11.7
AG	More than 100 DP seats	5	140 (140)	7.6
VS	More than 100 DP seats	3	130 (130)	2.1
GR	Context only (former Majorz)	2	120 (120)	3.8
SZ	At most 100 DP seats	3	100 (100)	4.0
ZG	At most 100 DP seats	3	80 (80)	2.9
NW	At most 100 DP seats	4	60 (60)	6.7
SH	At most 100 DP seats	5	60 (60)	15.7
UR	At most 100 DP seats	2	37 (64)	1.4

The mean of the canton-level gains is 7.1 percentage points in the group with more than 100 Double Pukelsheim seats and 6.1 percentage points in the group with at most 100. Their difference (more than 100 minus at most 100) is +1.0 percentage points. With only 8 cantons and distinct electoral rules, this is a descriptive comparison rather than evidence of a causal effect of the number of Double Pukelsheim seats.

Interpretation of the Results

Discussion

Conclusion

Bibliography

Amt für Statistik und Daten Kanton Zürich. (2026). *Archiv Wahlresultate Kantonsratswahlen* [Data set]. Kanton Zürich. https://www.zh.ch/de/politik-staat/wahlen-abstimmungen/wahlarchiv.zhweb-noredirect.zhweb-cache.html#/datasets/690@statistisches-amt-kanton-zuerich

Bundesgericht. (1965). *BGE 91 I 266*. https://relevancy.bger.ch/php/clir/http/index.php?highlight_docid=atf%3A%2F%2F91-I-266%3Ade&lang=de&type=show_document

Bundesgericht. (2002). *BGE 129 I 185*. https://relevancy.bger.ch/php/clir/http/index.php?highlight_docid=atf%3A%2F%2F129-I-185%3Ade&lang=de&type=show_document

Bundesgericht. (2005). *BGE 131 I 74*. https://servat.unibe.ch/dfr/bge/c1131074.html

Bundesgericht. (2010a). *BGE 136 I 352*. https://relevancy.bger.ch/php/clir/http/index.php?highlight_docid=atf%3A%2F%2F136-I-352%3Ade&lang=de&type=show_document

Bundesgericht. (2010b). *BGE 136 I 376*. https://search.bger.ch/ext/eurospider/live/de/php/clir/http/index.php?highlight_docid=atf%3A%2F%2F136-I-376%3Ade&lang=de&type=show_document&zoom=IN

Bundesgericht. (2012, 19. März). *Urteil 1C_407/2011, 1C_445/2011 und 1C_447/2011: Kantonsratswahlen 2012, Verletzung des Wahlrechts*. https://search.bger.ch/ext/eurospider/live/de/php/aza/http/index.php?highlight_docid=aza%3A%2F%2F19-03-2012-1C_407-2011&lang=de&type=show_document

Bundesgericht. (2013). *BGE 139 I 195*. https://relevancy.bger.ch/php/clir/http/index.php?highlight_docid=atf%3A%2F%2F139-I-195%3Ade&lang=de&type=show_document

Bundeskanzlei. (2013, August 21). *Proporzwahlsysteme im Vergleich: Bericht der Bundeskanzlei*. https://politchronik.swiss/references/175498/download?filename=Bericht+BK+vom+21.8.13_1151_20130821.pdf&inline=true&locale=de

Bundeskanzlei. (2025, June 6). *Wahlsysteme und Sitzzuteilungsverfahren im Vergleich: Bericht der Bundeskanzlei*. https://cms.news.admin.ch/fileservice/sdweb-docs-prod-nsbcch-files/files/2025/08/21/fdacd8e9-e27f-4e8d-a5f0-c3da45ed88ff.pdf

Department für Finanzen und Ressourcen des Kantons Aargau. (2009, April 21). *Detailergebnisse der Grossratswahlen vom 8. März 2009*. https://www.ag.ch/de/medien/medienmitteilungen?mm=detailergebnisse-der-grossratswahlen-vom-8-maerz-2009-742f7d21-d4b0-454f-abd7-25471ef7f566_de

Gilg, P. (2014, December 27). Wahlsysteme. In *Historisches Lexikon der Schweiz*. https://hls-dhs-dss.ch/de/articles/026454/2014-12-27/

Grünliberale Partei Kanton Zürich. n.d. *Unsere Geschichte*. Accessed September 15, 2026. https://zh.grunliberale.ch/partei/verein/unsere-geschichte/

Grosser Rat des Kantons Thurgau. (n.d.). *Kantonale Volksinitiative „Mehr Transparenz bei Wahlen im Thurgau“ (Doppelproporz nach Pukelsheim)*. Retrieved September 17, 2026, from https://parlament.tg.ch/de/geschaefte/detail.php?gid=97e7bade75ba4cadabae69f649c29b94

Initiativkomitee “Damit jede Stimme zählt.” (n.d.). *Initiative für ein faires Wahlsystem im Kanton St. Gallen*. Retrieved September 15, 2026, from https://jede-stimme-zaehlt.ch/

Kanton Aargau. (2004). *Grossratswahlgesetz: Änderung vom 8. Juni 2004*. Abstimmungsbroschüre zur kantonalen Volksabstimmung vom 26. September 2004, pp. 5–8. https://www.ag.ch/media/kanton-aargau/portal/aktuell/wahlen-abstimmungen/abstimmungen/abstimmungsbroschueren/20040926-abstimmungsbroschuere.pdf

Kanton Aargau. (2008). *Abstimmungsvorlagen vom 24. Februar 2008: Grossratswahlgesetz (Gesetz über die Wahl des Grossen Rates), Änderung vom 18. September 2007*, pp. 6–25. https://www.ag.ch/media/kanton-aargau/portal/aktuell/wahlen-abstimmungen/abstimmungen/abstimmungsbroschueren/20080224-abstimmungsbroschuere.pdf

Kanton Aargau. (2009). *Gesetz über die Wahl des Grossen Rates (Grossratswahlgesetz; Stand 1. Januar 2009)*. https://gesetzessammlungen.ag.ch/api/de/versions/519/pdf_file

Kanton Basel-Landschaft. (2025a). *Abstimmung vom 9. Februar 2025: Abstimmungsvorlagen* [Abstimmungsbroschüre]. https://abstimmungen.bl.ch/app/archive/de/storage/cd604418d5ac943f3373dd673738f3101f41769264c57eba9d692ea3d9a12347.pdf

Kanton Basel-Landschaft. (2025b). *Einführung des kantonalen Doppelproporzes im Kanton Basel-Landschaft – Teilrevision des Gesetzes über die politischen Rechte* [Abstimmungsresultat vom 9. Februar 2025]. https://abstimmungen.bl.ch/app/archive/de/vote/ct-13-100.html

Kanton Graubünden. (2022). *Pressedaten Proporzwahl kantonal: Grossratswahlen 2022* [Excel spreadsheet]. https://www.gr.ch/DE/publikationen/abstimmungenwahlen/Grossratswahlen-2022/ResultateGrossratswahlen/20220515_Pressedaten_Proporzwahl_kantonal.xlsx

Kanton Graubünden. (2026). *Pressedaten Proporzwahl kantonal: Grossratswahlen 2026* [Excel spreadsheet]. https://www.gr.ch/DE/publikationen/abstimmungenwahlen/grw26/Resultate1/20260614_Pressedaten_Proporzwahl_kantonal.xlsx

Kanton Schaffhausen. (n.d.). *Frühere Urnengänge (seit 2003)*. https://archiv.sh.ch/daten/Fruehere-Urnengaenge-seit-200.1127.0.html

Kanton Schwyz. (2024). *Protokoll der Erneuerungswahl Kantonsrat 2024 vom 3. März 2024*. https://kanzlei.sz.ch/data/publication/issues/67/downloads/de/20240303_Wahlprotokoll_Gemeinde%C3%BCbersicht_Proporzwahl_kantonal.pdf

Kanton Uri. (2018). *Synoptische Darstellung zum Gesetz über die Verhältniswahl des Landrats (Proporzgesetz)*. https://www.ur.ch/_docn/113506/LA.2017-0413_IV._Synoptische_Darstellung_Proporzgesetz.pdf

Kanton Wallis. (2004a). *Gesetz über die politischen Rechte (kGPR) vom 13. Mai 2004* (SGS 160.1; Stand 9. April 2017). https://lex.vs.ch/app/de/texts_of_law/160.1/versions/1375

Kanton Wallis. (2016, 29. September). *Legislaturperiode 2017–2021: Verteilung der Anzahl Grossratssitze pro Bezirk*. https://www.vs.ch/documents/529400/2263784/MM_Verteilung_Grossratssitze.pdf/30be6442-6e9e-4d91-adef-76d3f60d6107

Kanton Wallis. (2020, 25. September). *Legislaturperiode 2021–2025: Verteilung der Anzahl Grossratssitze pro Bezirk*. https://www.vs.ch/documents/529400/8932772/2020+09+25+-+MM+-+Verteilung+Grossratssitze.pdf/62d59d10-2075-5231-e385-88688c7cb559

Kanton Wallis. (2024, 20. September). *Anhang zur Verteilung der Anzahl Grossratssitze pro Bezirk, Legislaturperiode 2025–2029*. https://www.vs.ch/documents/529400/33671442/Anhang+Verteilung.pdf/b1f92f33-f4b2-35ef-6149-838af6c70b09

Kanton Zürich. (1983/2005). *Gesetz über die Wahlen und Abstimmungen (Wahlgesetz)* (LS 161; Nachtrag 47, Stand 1. Januar 2005), §§ 87–88b. https://www.notes.zh.ch/appl/zhlex_r.nsf/WebView/C4AB55033822301DC1256F4A0047A916/$File/161_4.9.83_47.pdf

Kanton Zug. (2006, September 28). *Gesetz über die Wahlen und Abstimmungen (Wahl- und Abstimmungsgesetz, WAG)* (Vorlage Nr. 1300.24, Laufnummer 12 198). https://kr-geschaefte.zug.ch/dokumente/2793/pdoc_1494_1.pdf

Kanton Zug. (2022a, July 1). *Kantonsratswahl 2018* [Data set]. https://wab.zug.ch/elections/kantonsratswahl-2018/data

Kanton Zug. (2022b, July 1). *Kantonsratswahl Kanton Zug* [Data set]. https://wab.zug.ch/elections/kantonsratswahl-kanton-zug-2014/data

Kanton Zug. (2022c, October 2). *Kantonsratswahl 2022* [Data set]. https://wab.zug.ch/elections/kantonsratswahl-2022/data

Laakso, M., & Taagepera, R. (1979). "Effective" number of parties: A measure with application to West Europe. *Comparative Political Studies, 12*(1), 3–27. https://doi.org/10.1177/001041407901200101

Leuzinger, L. (2018). Die Auswirkungen des Doppelproporz bei kantonalen Parlamentswahlen. *Parlament, Parlement, Parlamento, 21*(3), 38–47. https://pemmenegger.com/wp-content/uploads/2020/12/181122_buch_parlament_03_2018_pe-v2.pdf

Lüscher, S. (2023). Parlamentswahlen nach dem doppelt-proportionalen Sitzzuteilungsverfahren im Lichte der territorialen Repräsentation: Eine empirische Annäherung an ein konzeptionelles Dilemma. *Zeitschrift für Parlamentsfragen, 54*(3), 571–595. https://doi.org/10.5771/0340-1758-2023-3-571

Lüscher, S. (2024). *Redeeming the pledge of electoral equity? A study of the roads to and effects of the divisor-based biproportional apportionment method in Switzerland* [Doctoral dissertation, University of Zurich]. https://doi.org/10.5167/uzh-264856

Lüscher, S. (2025, February 20). *20 Jahre Doppelproporz—eine Rückschau*. DeFacto. https://www.zora.uzh.ch/entities/publication/9b83ccf9-a61f-45ff-b0c8-c52b5cfcb1c7

Pukelsheim, F., & Schuhmacher, C. (2011). Doppelproporz bei Parlamentswahlen – ein Rück- und Ausblick. *Aktuelle Juristische Praxis / Pratique Juridique Actuelle, 20*(12), 1581–1599. https://www.math.uni-augsburg.de/htdocs/emeriti/pukelsheim/2011e.pdf

Regierungsrat des Kantons Aargau. (2007, July 4). *Revision der Kantonsverfassung und des Grossratswahlgesetzes (Gesetz über die Wahl des Grossen Rates): Bericht und Entwurf zur 2. Beratung (Botschaft 07.176)*. https://www.ag.ch/media/kanton-aargau/portal/aktuell/wahlen-abstimmungen/abstimmungen/archiv/2008-02-24/v3-und-4-botschaft-rr-2-beratung.pdf

Regierungsrat des Kantons Aargau. (2013). *Jahresbericht mit Jahresrechnung 2012. Kanton Aargau*. https://www.ag.ch/media/kanton-aargau/dfr/dokumente/finanzen/planung-und-berichterstattung/jahresbericht-mit-jahresrechnung/botschaft-2012.pdf

Regierungsrat des Kantons Luzern. (2026, March 31). *Volksinitiativen «Faire Wahlen ohne Listenflut» und «Stopp Listenflut»: Botschaft des Regierungsrates an den Kantonsrat* (B 93). https://www.lu.ch/-/klu/ris/cdws/document?fileid=641a78c026024e88bd438d2f0d3d0462

Regierungsrat des Kantons Schaffhausen. (2007, April 24). *Bericht und Antrag des Regierungsrates des Kantons Schaffhausen an den Kantonsrat betreffend Anpassung des Wahlsystems für den Kantonsrat (Bericht und Antrag 07-41)*. https://archiv.sh.ch/daten/fileadmin/Redaktoren/Dokumente_nicht_im_Formularpool/Regierung/Vorlagen/2007/24-4-07.pdf

Regierungsrat des Kantons Zürich. (2014, November 26). *Postulat (Änderung Wahlgesetz [Wahlkreiseinteilung]) (RRB Nr. 1254/2014; KR-Nr. 219/2014)*. https://parlzhcdws.cmicloud.ch/parlzh5/cdws//Files/89dffabbc7a240f08e824f7a30bebf4d-332/1/pdf

Regierungsrat des Kantons Zürich. (2024, March 27). *Gesetz über die politischen Rechte (GPR): Anpassung der Regelung zum 3%-Quorum bei Kantonsratswahlen* (Vorlage 5951). https://www.zh.ch/bin/zhweb/publish/regierungsratsbeschluss-unterlagen./2024/338/5951_GPR_Aenderung.pdf

Regierungsrat des Kantons Zürich. (2025, October 1). Gesetz über die politischen Rechte: Anpassung der Regelung zum 3%-Quorum bei Kantonsratswahlen, Inkraftsetzung (Regierungsratsbeschluss Nr. 998/2025). https://www.zh.ch/bin/zhweb/publish/regierungsratsbeschluss-unterlagen./2025/998/RRB-2025-0998.pdf

Staat Wallis. (2017). *Grossräte 2017 – Resultate: Offizielle Endresultate. Archivierte Abstimmungs- und Wahlresultate*. https://vework-public.vs.ch/app/archive/sites/votel2/DE/1/result/gc/137

Staatskanzlei des Kantons Aargau. (2005, March 7). *Bezirksprotokolle über die Erneuerungswahl des Grossen Rats für die Amtsperiode 2005/2009: Wahltag: 27. Februar 2005 (Amtsblattbeilage 2005-1)*. https://doczz.net/doc/5954970/amtsblattbeilage_vom_07.03.2005

Staatskanzlei des Kantons Aargau. (2008, September 15). *Grossratswahlen: Vorverfahren gestartet*. https://www.ag.ch/de/medien/medienmitteilungen?mm=Grossratswahlen-Vorverfahren-gestartet-50a84551-3d6d-454c-a33c-5644c8bf50a5_de

Staatskanzlei des Kantons Aargau. (2012, April 5). *Gemeinsame Grossrats- und Regierungsratswahlen: Vorverfahren gestartet: Anmeldefristen sind bekannt*. https://www.ag.ch/de/medien/medienmitteilungen?mm=gemeinsame-grossrats-und-regierungsratswahlen-e7d48210-c7f0-46ac-8a70-ffc1aab7b4d6_de

Staatskanzlei des Kantons Aargau. (2016, March 18). *Ausschreibung der Grossrats- und Regierungsratswahlen 2016: Unterlagen unter www.ag.ch/wabag verfügbar*. https://www.ag.ch/de/medien/medienmitteilungen?mm=ausschreibung-der-grossrats-und-regierungsratswahlen-2016-f7404346-a5eb-46e0-8627-301716c82855_de

Staatskanzlei Nidwalden. (2014). *Wahlergebnis der Landratswahlen vom 23. März 2014* [Unpublished raw data]. Kanton Nidwalden.

Staatskanzlei Nidwalden. (2018). *Wahlergebnis der Landratswahlen vom 4. März 2018* [Unpublished raw data]. Kanton Nidwalden.

Staatskanzlei Nidwalden. (2022). *Wahlergebnis der Landratswahlen vom 13. März 2022* [Unpublished raw data]. Kanton Nidwalden.

Staatskanzlei Nidwalden. (2026). *Wahlergebnis der Landratswahlen vom 8. März 2026* [Unpublished raw data]. Kanton Nidwalden.

Staatskanzlei Kanton Schaffhausen. (2020, September 27). *Resultate der Kantonsratswahl vom 27. September 2020*. Kanton Schaffhausen. https://sh.ch/CMS/Webseite/Kanton-Schaffhausen/Beh-rde/Abstimmungen-und-Wahlen-6275841-DE.html

Staatskanzlei Kanton Schaffhausen. (2024, September 22). *Kantonsratswahlen vom 22. September 2024 – Resultate*. Kanton Schaffhausen. https://www.sh.ch/CMS/Webseite/Kanton-Schaffhausen/Beh-rde/Regierung/Staatskanzlei-15401466-DE.html

Standeskanzlei Uri. (2020). *Landratswahlen 2020: Statistische Auswertung*. Kanton Uri. https://www.ur.ch/_docn/213898/Landratswahlen_2020_stat_Auswertung.pdf

Standeskanzlei Uri. (2024a). *AuswertungKapitel2-2024* [Unpublished Excel workbook]. Kanton Uri.

Standeskanzlei Uri. (2024b). *Landratswahlen 2024: Statistische Auswertung*. Kanton Uri. https://www.ur.ch/_docn/384232/Landratswahlen_2024_stat_Auswertung.pdf

Statistisches Amt Kanton Aargau. (n.d.). *Grossratswahlen: Detailresultate, ab 1985* [Data set]. Kanton Aargau. Retrieved September 1, 2026, from https://www.ag.ch/app/staag/statistikportal/

Statistisches Amt des Kantons Zürich. 2003. *Die Zürcher Kantonsratswahlen 2003: Anhang Tabelle 3, Listen und Listenverbindungen.* Microsoft Excel spreadsheet. Accessed September 15, 2026. https://web.archive.org/web/20031207213505id_/http://www.statistik.zh.ch/statistik.info/xls/2003_23.xls.

Statistisches Amt des Kantons Zürich. (2011). *Statistisches Jahrbuch des Kantons Zürich 2011.* https://www.web.statistik.zh.ch/ogd/daten/ressourcen/KTZH_00000124_00001234.pdf

Appendix

Appendix 1: Explanation Data Cleaning Process

Appendix 2: Code for Data Cleaning

Declaration of AI-Assistance